Question 1

Create an array A with elements `(12, 13, 14, 15, 16)` and display them.

```
A <- c(12, 13, 14, 15, 16)
A
```

Output

```
[1] 12 13 14 15 16
```

Question 2

Find the sum of all elements of `A`

```
sum(A)
```

Output

```
[1] 70
```

Question 3

Find the product of all elements of `A`

```
prod(A)
```

Output

```
[1] 524160
```

Question 4

Find the maximum and minimum element of `A`

```
max(A)
min(A)
```

Output

```
[1] 16[1] 12
```

Question 5

Find the range of array `A`

```
range(A)
```

Output

```
[1] 12 16
```

Question 6

Find the mean, variance and standard deviation and median of values of `A`

```
mean(A)
var(A)
sd(A)
```

Output

```
[1] 14[1] 2.5[1] 1.581139
```

Question 7

Sort the elements of A in both increasing and decreasing order and store them in `B` and `C`

```
B <- sort(A)
B

C <- sort(A, decreasing=TRUE)
C
```

Output

```
[1] 12 13 14 15 16[1] 16 15 14 13 12
```

Question 8

Create a matrix of 3x4 to have the set of natural numbers

```
mat <- matrix(seq(1, 12), nrow=3, ncol=4)
print(mat)
```

Output

```
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12
```

Question 9

Create MxN matrix by combining A, B and C row-wise (RW) and column-wise (CW).

```
rw <- rbind(A, B, C)
print(rw)

cw <- cbind(A, B, C)
print(cw)
```

Output

```
  [,1] [,2] [,3] [,4] [,5]
A   12   13   14   15   16
B   12   13   14   15   16
C   16   15   14   13   12
      A  B  C
[1,] 12 12 16
[2,] 13 13 15
[3,] 14 14 14
[4,] 15 15 13
[5,] 16 16 12
```

Question 10

Find the 10 and 3 row elements of RW

```
print(rw[10])
print(rw[3])
```

Output

```
[1] 15
[1] 16
```

Question 11

Find 1 and 4 columns of CW

```
print(cw[1])
print(cw[4])
```

Output

```
[1] 12
[1] 15
```

Question 12

Using both RW and CW find sub-matrices having elements `[2, 3]` and `[2, 4]`

```
print("First Matrix")
print(rw[2, 3])

print("Second Matrix")
print(rw[2, 4])
```

Output

```
[1] "First Matrix"
 B
14
[1] "Second Matrix"
 B
15
```

```
print("First Matrix")
print(cw[2, 3])

print("Second Matrix")
print(cw[2, 4])
```

Output

```
[1] "First Matrix"
 C
15
[1] "Second Matrix"
```